Euclid's Elements — Book I
Proposition 32

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In any triangle, if one of the sides is produced, the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.

1 Introduction

Proposition I.32 is the first proposition in Book I that applies the Euclidean parallel theory developed in I.29–I.31 directly to a triangle.

Let ABC be a nondegenerate triangle and extend BC through C to D. Euclid asserts first that the exterior angle at C is equal to the two remote interior angles:

$$\angle ACD = \angle BAC + \angle ABC.$$

He then concludes that the three interior angles of the triangle are together equal to two right angles.

In the present reconstruction no numerical angle measure is introduced. Thus the familiar modern formula

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

is not the literal Lean statement. Instead the proposition is represented synthetically: a ray inside the exterior angle decomposes it into two angles congruent to the two remote interior angles, while $B - C - D$ records the straight-angle configuration at C.

This distinction is mathematically important. The formal proof establishes the synthetic content of the 180° theorem without introducing degrees or an arithmetic operation of angle addition.

2 Classical Statement

Let ABC be a triangle and let BC be produced to D. Proposition I.32 has two conclusions. First,

$$\angle ACD = \angle BAC + \angle ABC.$$

Second,

$$\angle BAC + \angle ABC + \angle ACB = 2R,$$

where $2R$ denotes two right angles.

The classical Euclidean proof constructs through C a line parallel to AB. Proposition I.29 then identifies the two parts of the exterior angle with the two remote interior angles.

3 The Formal Configuration

The Lean proof refines the classical diagram because the formal argument must record the orientation of rays and the incidence data needed by the parallel-angle theorem.

The principal points satisfy

$$A - M - C, \qquad B - M - E, \qquad A - R - D, \qquad B - C - D.$$

The auxiliary construction gives

$$AB \parallel CE.$$

Moreover E and R determine the same ray from C. Thus the ray CR lies inside the exterior angle ACD and decomposes it into the two angles used in the formal statement.

For the explicit application of Proposition I.29 the proof also extends the two parallel lines:

$$A - B - X, \qquad E - C - F.$$

Hence

$$AX \parallel EF.$$

4 Synthetic Representation of the Exterior-Angle Equality

The first conclusion is represented by the predicate

```
HilbertExteriorAngleEqualsRemoteAngles
```

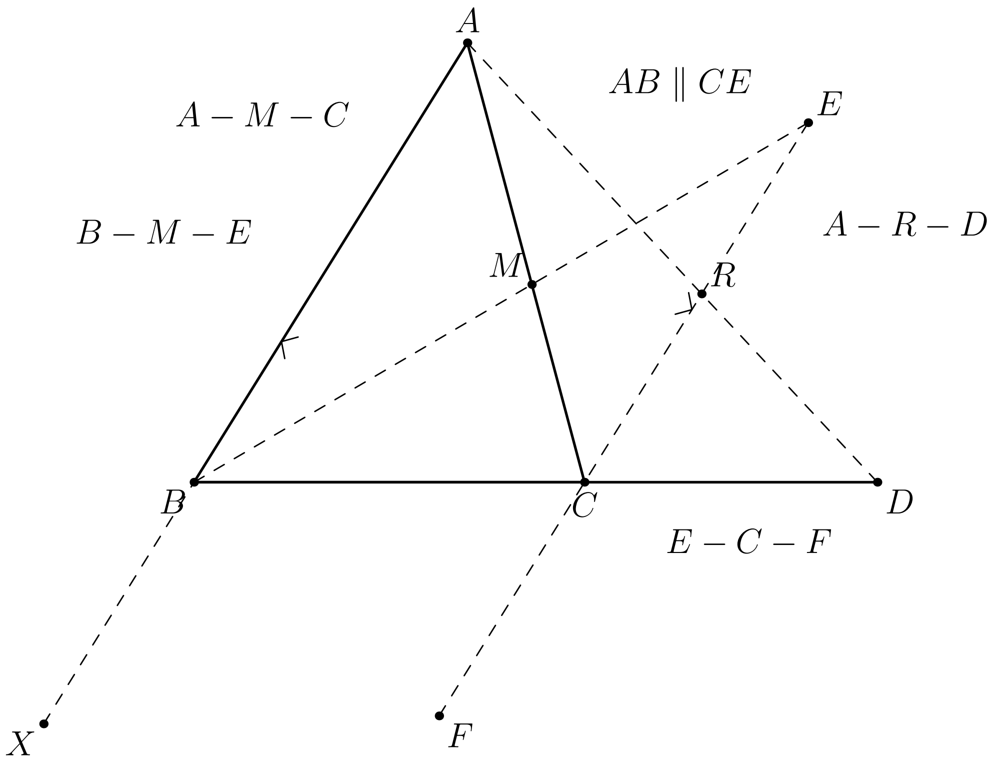

Figure 1: The formal configuration of Proposition I.32. The essential parallel relation is $AB \parallel CE$. The points X and F are technical extensions used to put the configuration into the explicit transversal form of Proposition I.29.

defined by

```
def HilbertExteriorAngleEqualsRemoteAngles
    [HilbertIncidence Geo]
    (A B C D : Geo.Point) : Prop :=
  exists R : Geo.Point,
    Geo.Between A R D /\
    Geo.AngleCongruent B A C A C R /\
    Geo.AngleCongruent A B C R C D
```

Thus the equation

$$\angle ACD = \angle BAC + \angle ABC$$

is encoded without angle addition. Instead there exists a point R with

$$A - R - D$$

such that

$$\angle BAC \cong \angle ACR$$

and

$$\angle ABC \cong \angle RCD.$$

The ray CR is therefore the synthetic "division point" of the exterior angle.

5 Construction of the Correctly Oriented Parallel Ray

A naive use of Proposition I.31 would produce a line through C parallel to AB, but a line alone does not determine which of its two rays lies inside the exterior angle ACD.

This orientation issue is genuine in the formal proof.

The reconstruction therefore uses the already established auxiliary configuration

```
hilbert_exterior_angle_aux
```

which supplies points M, E satisfying

$$A - M - C, \qquad B - M - E,$$

together with

$$\angle BAC \cong \angle MCE.$$

The theorem

```
hilbert_exterior_angle_aux_parallel
```

then yields

$$AB \parallel CE.$$

The important point is that E is not an arbitrary point on the parallel through C. Its orientation is inherited from the auxiliary exterior-angle construction.

The helper

```
hilbert_exterior_angle_inside
```

uses this oriented configuration, together with $B - C - D$, to show that the ray CE meets the open segment AD. Hence there is a point R such that

$$A - R - D$$

and E, R lie on the same ray from C.

This is the formal crossbar step of the proof.

6 First Remote Interior Angle

From

$$A - M - C$$

the points M and A lie on the same ray from C. Therefore

$$\angle MCE = \angle ACE.$$

Since E and R lie on the same ray from C,

$$\angle ACE = \angle ACR.$$

Combining these ray transports with the congruence supplied by the auxiliary construction gives

$$\boxed{\angle BAC \cong \angle ACR}.$$

In Lean this is the intermediate result

```
hFirst :
    Geo.AngleCongruent B A C A C R
```

and it establishes the first half of the exterior-angle decomposition.

7 Preparing Proposition I.29

The second congruence is obtained from the parallel relation, but the explicit theorem

```
euclid_proposition_29_transversal
```

expects the two intersections to lie between endpoints of the parallel lines.
The proof therefore extends AB beyond B:

$$A - B - X,$$

and extends EC beyond C:

$$E - C - F.$$

Parallelism is transported along the corresponding collinear triples to obtain

$$AX \parallel EF.$$

The transversal is the line BD, with intersection points B and C.

There remains an orientation condition. Since $A - R - D$, the points A and R lie on the same side of BD. From the same-ray relation between E and R, together with $E - C - F$, the proof obtains

$$R - C - F.$$

Thus R and F lie on opposite sides of BD. Transporting this relation through the same-side relation between A and R gives

$$A \text{ and } F \quad \text{on opposite sides of } BD.$$

At this point all hypotheses of the explicit transversal form of I.29 are available.

8 Second Remote Interior Angle

Proposition I.29 applied to

$$AX \parallel EF$$

with transversal BD gives

$$\angle ABC \cong \angle BCF.$$

At C the two straight lines are represented by

$$B - C - D \qquad \text{and} \qquad E - C - F.$$

Proposition I.15, through the theorem `VerticalAngles`, therefore gives

$$\angle BCF \cong \angle ECD.$$

By transitivity,

$$\angle ABC \cong \angle ECD.$$

Finally E and R lie on the same ray from C, so the first arm of the second angle may be transported from E to R:

$$\boxed{\angle ABC \cong \angle RCD}.$$

This is the Lean result

```
hSecond :
    Geo.AngleCongruent A B C R C D
```

and completes the exterior-angle theorem.

9 Proof Architecture of the Exterior-Angle Part

The actual formal dependency structure is

$$\begin{array}{c} ABC \text{ noncollinear}, \quad B - C - D \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_aux} \\ \downarrow \\ A - M - C, \quad B - M - E, \quad \angle BAC \cong \angle MCE \\ \downarrow \\ AB \parallel CE \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_inside} \\ \downarrow \\ A - R - D, \quad \mathrm{SameRay}(C, E, R) \end{array}$$

and then the proof splits:

$$\begin{array}{ccc} & A - R - D, \quad AB \parallel CE & \\ \swarrow & & \searrow \\ \text{same-ray transport} & & \text{extend the two parallels} \\ \downarrow & & \downarrow \\ \angle BAC \cong \angle ACR & & AX \parallel EF \\ & & \downarrow \\ & & \text{I.29 + vertical angles} \\ & & \downarrow \\ & & \angle ABC \cong \angle RCD. \end{array}$$

The two branches are then packaged as

$$\exists R, \quad A - R - D, \quad \angle BAC \cong \angle ACR, \quad \angle ABC \cong \angle RCD.$$

10 Where the Euclidean Parallel Axiom Enters

The construction and order-theoretic parts of the proof use the Hilbert incidence, order, congruence, side-of-line and crossbar machinery.

The specifically Euclidean step is the use of Proposition I.29:

$$AX \parallel EF \longrightarrow \angle ABC \cong \angle BCF.$$

As established in the reconstruction of I.29, this direction depends on Hilbert's Group IV, or equivalently on uniqueness of the parallel through a point.

Thus I.32 inherits its Euclidean character from I.29.

This is structurally different from I.27 and I.28, where angle conditions imply parallelism in neutral geometry. In I.32 the proof needs the converse direction:

$$\text{parallelism} \longrightarrow \text{angle congruence.}$$

11 The Triangle-Angle-Sum Statement

The second formal predicate is

```
HilbertTriangleAnglesEqualTwoRightAngles
```

defined by

```
def HilbertTriangleAnglesEqualTwoRightAngles
    [HilbertIncidence Geo]
    (A B C : Geo.Point) : Prop :=
  exists D : Geo.Point,
    Geo.Between B C D /\
    HilbertExteriorAngleEqualsRemoteAngles
      A B C D
```

The theorem

```
euclid_proposition_32
```

is consequently short. From noncollinearity one obtains $B \neq C$, extends BC through C to a point D, and applies

```
euclid_proposition_32_exterior
```

to that extension.

The proof architecture is simply

$$\begin{gathered} ABC \text{ noncollinear} \\ \downarrow \\ B \neq C \\ \downarrow \\ \text{extend } BC \text{ through } C \\ \downarrow \\ B - C - D \\ \downarrow \\ \texttt{euclid\_proposition\_32\_exterior} \\ \downarrow \\ \texttt{HilbertTriangleAnglesEqualTwoRightAngles}. \end{gathered}$$

12 What "Equal to Two Right Angles" Means Here

There is an important distinction between the classical mathematical consequence and the literal formal statement.

In ordinary angle-measure notation, I.32 yields

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ.$$

The present Hilbert layer, however, does not assign numerical measures to angles. Nor does it define a general arithmetic operation

$$\alpha + \beta + \gamma.$$

Instead the formalization records the synthetic configuration from which the two-right-angles conclusion follows:

$$B - C - D$$

makes the rays CB and CD opposite, so $\angle ACB$ and $\angle ACD$ form a straight-angle configuration. The exterior angle is then decomposed by CR into angles congruent to $\angle BAC$ and $\angle ABC$.

Consequently the formal theorem proves the geometric content

$$\angle BAC + \angle ABC + \angle ACB = 2R$$

without translating $2R$ into the numerical value 180°.

Strictly speaking, the current predicate `HilbertTriangleAnglesEqualTwoRightAngles` stores the extension $B-C-D$ and the exterior-angle decomposition; the supplementary interpretation of the straight line at C is implicit in this data rather than represented by a separate angle-addition object.

This is not merely a coding detail. It identifies exactly what must be formalized in a synthetic geometry before the familiar numerical 180° formulation can even be stated.

13 Lean Formalization

The principal theorem is

```
euclid_proposition_32_exterior
```

with hypotheses

```
(A B C D : Geo.Point)
(hABC : Not (PrimCollinear Geo A B C))
(hBCD : Geo.Between B C D)
```

and conclusion

```
HilbertExteriorAngleEqualsRemoteAngles A B C D
```

The proof uses the following main components:

```
hilbert_exterior_angle_aux
hilbert_exterior_angle_aux_parallel
hilbert_exterior_angle_inside
hilbert_angle_eq_of_sameRay_first
hilbert_angle_eq_of_sameRay_second
hilbert_between_points_sameSide_transversal
euclid_proposition_29_transversal
VerticalAngles
```

The second theorem,

```
euclid_proposition_32
```

is a corollary obtained by extending BC and invoking the exterior-angle theorem.

The typeclass

```
[HilbertEuclideanPlane Geo]
```

appears because the proof ultimately uses Proposition I.29 and hence the Euclidean parallel axiom.

14 Relation to the Classical Proof

At the level of a paper proof, I.32 is short:

$$\text{construct a parallel through } C \longrightarrow \text{use I.29 twice} \longrightarrow \text{add the adjacent angles.}$$

The Lean reconstruction exposes several pieces hidden by this description.

First, a parallel *line* through C is insufficient: the correct orientation of the ray must be established.

Second, the ray used to split the exterior angle must be shown to meet the segment AD. This is the crossbar component.

Third, the explicit I.29 theorem carries side-of-line hypotheses that must be proved from the betweenness configuration.

Fourth, the proof does not "add angles" numerically. It records a synthetic decomposition of the exterior angle.

Thus the expanded formal proof is not evidence that Euclid's argument is mathematically complicated. It is an audit of the incidence, order and orientation information suppressed by the classical diagram.

15 Conclusion

Proposition I.32 combines the parallel theory of I.29–I.31 with the synthetic theory of triangles.

Its essential Euclidean step is

$$AB \parallel CE \longrightarrow \text{the required angle congruence},$$

which is supplied by I.29.

The formal proof then shows that the correctly oriented parallel ray through C cuts the exterior angle into two parts congruent to the remote interior angles:

$$\angle BAC \cong \angle ACR, \qquad \angle ABC \cong \angle RCD.$$

Together with the straight-line relation $B - C - D$, this is the synthetic content of the classical theorem

$$\boxed{\angle BAC + \angle ABC + \angle ACB = 180^\circ}.$$

The reconstruction therefore reaches the familiar 180° theorem without introducing degrees: the numerical formulation appears only as the modern interpretation of a purely synthetic Hilbert-style configuration.